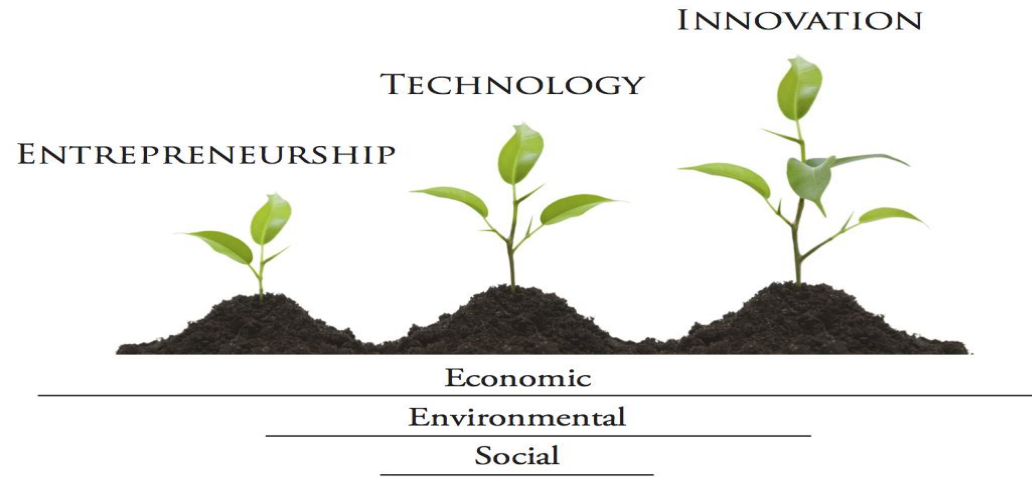




Entrepreneurship in IT

Ruba Awayes

Developing Successful Business Ideas

Recognizing opportunities and generating ideas

What is an idea?

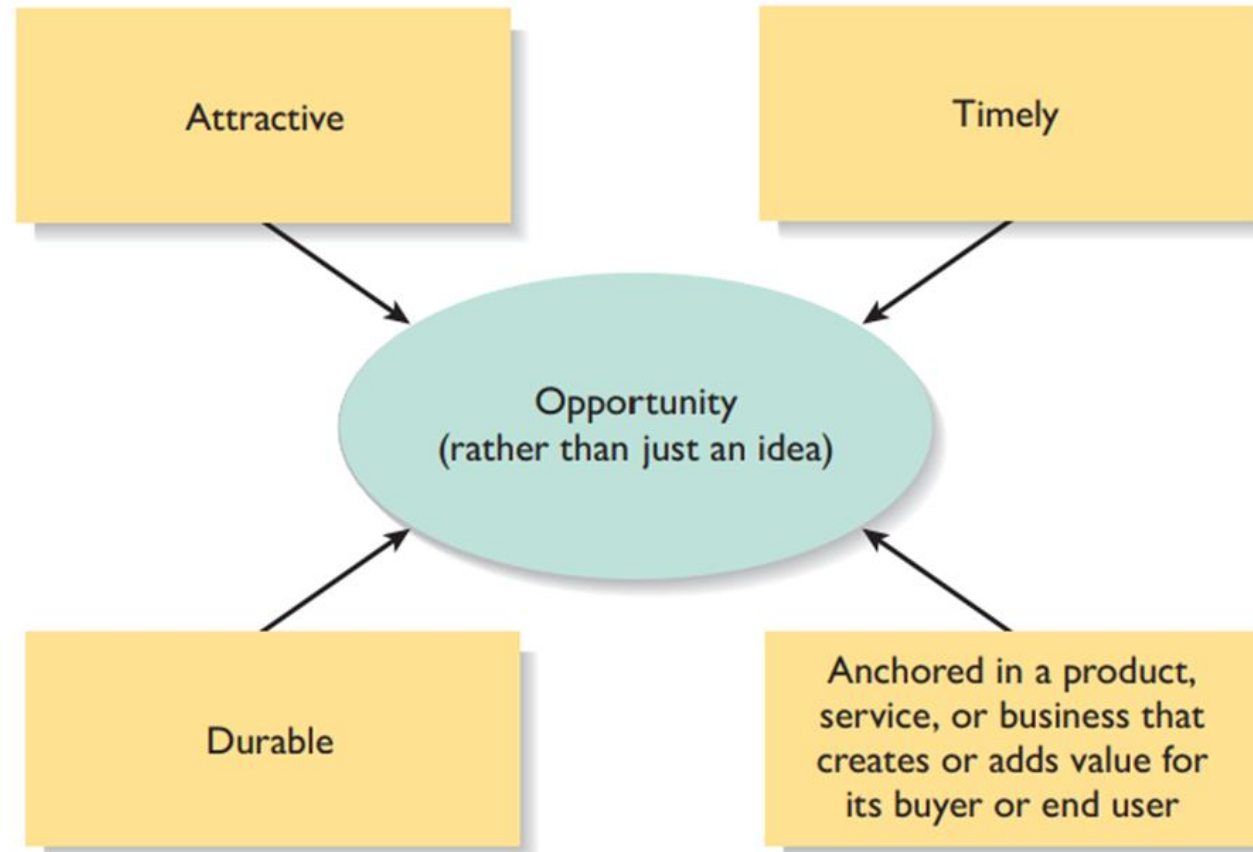
- An idea is a thought, an impression, or a notion that may or may not meet the criteria of an opportunity.
 - Before getting excited about a business idea, it is crucial to understand if the idea fills a need and satisfies the criteria for an opportunity.

What is an Opportunity?

An opportunity is a favorable set of circumstances that creates a need for a new product, service, or business.

- Entrepreneurs recognize opportunity gaps and create businesses to fill them.
 - For an entrepreneur to capitalize on an opportunity, its window of opportunity must be open.

Four Essential Qualities of an Opportunity



Three Ways to Identify an Opportunity

Observing Trends

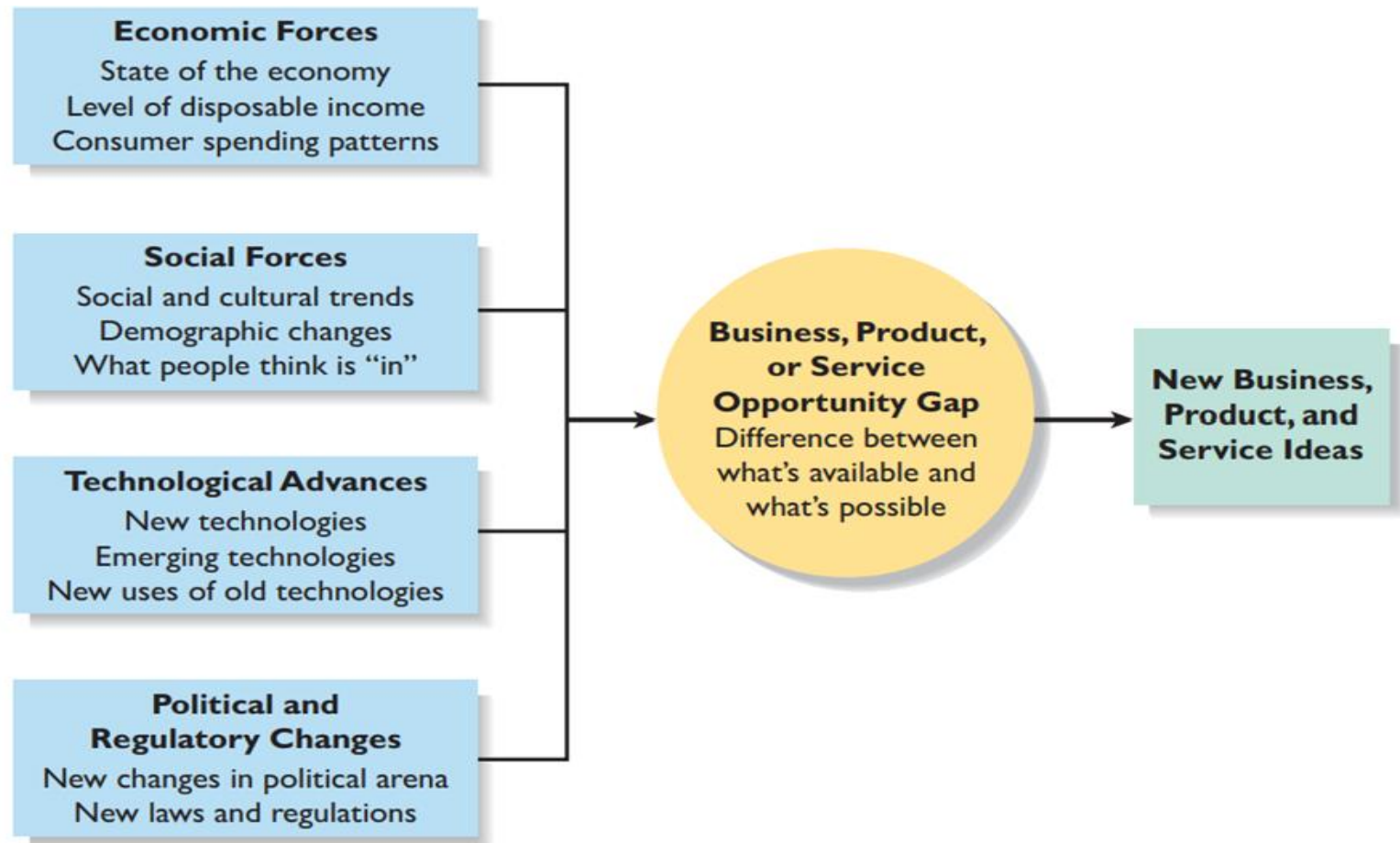
Solving a Problem

Finding Gaps in the
Marketplace

First Approach: Observing Trends

- Trends create opportunities for entrepreneurs to pursue.
- The most important trends are:
 - Economic forces
 - Social forces
 - Technological advances
 - Political and regulatory changes
- It's important to be aware of changes in these areas.

Environmental Trends Suggesting Business, Product, or Service Opportunity Gaps



Trend 1: Economic Forces

Economic trends help determine areas that are ripe for new start-ups and areas that start-ups should avoid.

Example of Economic Trend Creating a Favorable Opportunity

- A weak economy favors start-ups that help consumers save money.
 - An example is [GasBuddy.com](https://www.gasbuddy.com), a company started to help consumers save money on gas.
- The emergence of the creator economy is an important economic trend.

Trend 2: Social Forces

Social trends alter how people and businesses behave and set their priorities.

- **Examples of Social Trends**

- Aging of the population
- The increasing diversity of the population
- Shifting attitudes and buying behaviors of Gen Z
- Emphasis on sustainability
- A focus on physical fitness, mental health, and wellness
- Desire for connection to people and brands
- Desire for more flexible work schedules

Trend 3: Technological Advances

Advances in technology frequently create business opportunities.

- [OpenTable.com](https://opentable.com) helps people perform an everyday task in a more convenient and expedient manner.
- Artificial intelligence is helping startups provide enhanced products and services.
- Entire industries have been created as the result of technological advances.
 - Compatible products for Apple iPhone and Apple Watch

Trend 4: Political Action and Regulatory Changes (1 of 2)

Political and regulatory changes also provide the basis for opportunities.

General Example

This happened with the passage of the Affordable Care Act in 2010. The provisions of the act yielded opportunities for entrepreneurs to launch electronic records start-ups, apps to help patients monitor their medication, and similar companies.

Trend 4: Political Action and Regulatory Changes (2 of 2)

Political change also engenders new business and product opportunities.

- For example, global political instability and the threat of terrorism have resulted in many firms becoming more security conscious.

Cybersecurity is another threat for many nations and companies competing within them.

- Claroty is a cybersecurity startup providing its clients with a full range of cybersecurity controls for visibility, threat detection, risk, and vulnerability.

Second Approach: Solving a Problem

- Sometimes identifying opportunities simply involves noticing a problem and finding a way to solve it.
- These problems can be pinpointed through observing trends and through more simple means, such as intuition (حدس), serendipity (صدفة), or chance.
- Many companies have been started by people who have experienced a problem in their own lives, and then realized that the solution to the problem represented a business opportunity.
 - Java Jacket

A problem facing the world is finding alternatives to fossil fuels (الوقود الأحفوري)

A large number of entrepreneurial firms, like wind farm and solar energy are being launched to solve this problem.



Third Approach: Finding Gaps in the Marketplace (1 of 2)

- A gap in the marketplace is often created when a product or service is needed by a specific group of people but doesn't represent a large enough market to be of interest to mainstream retailers or manufacturers.
 - Ex. Daisy Rock Guitars.

Third Approach: Finding Gaps in the Marketplace (2 of 2)

- Many entrepreneurs realized there were no guitars on the market made specifically for females.

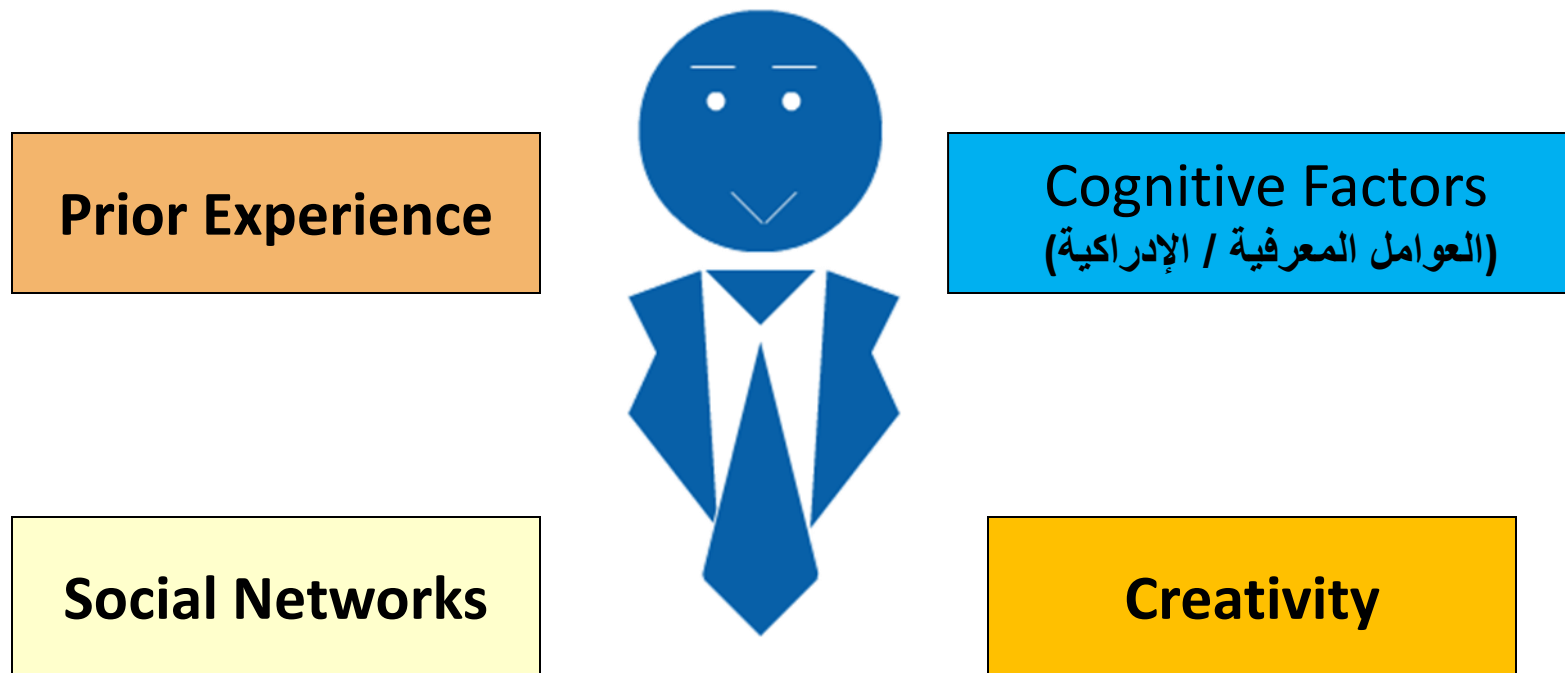


THE STANDARD SERIES
Reimagined. Refined. Remarkable.



- Chairs for left hand people

Personal Characteristics of the Entrepreneur



Prior Industry Experience

- Prior Industry Experience
 - Studies have shown that prior experience in an industry helps an entrepreneur recognize business opportunities.
 - By working in an industry, an individual may spot a market niche that is underserved.
 - It is also possible that by working in an industry, an individual builds a network of social contacts who provide insights that lead to recognizing new opportunities.

Cognitive Factors

- Cognitive Factors
 - Studies have shown that opportunity recognition may be an innate skill or cognitive process.
 - Some people believe that entrepreneurs have a “sixth sense” that allows them to see opportunities that others miss.
 - This “sixth sense” is called entrepreneurial alertness (يقظة ريادية), which is formally defined as the ability to notice things without engaging in deliberate search (بحث متعمد) .

Social Networks (1 of 2)

- Social Networks
 - The extent and depth of an individual's social network affects opportunity recognition.
 - People who build a substantial network of social and professional contacts will be exposed to more opportunities and ideas than people with sparse networks.
 - Research results suggest that between 40% and 50% of people who start a business got their idea via a social contact.
- Strong-Tie Vs. Weak-Tie Relationships
 - All of us have relationships with other people that are called “ties.”

Social Networks (2 of 2)

- Nature of Strong-Tie Vs. Weak-Tie Relationships
 - Strong-tie relationships are characterized by frequent interaction and form between coworkers, friends, and spouses.
 - Weak-tie relationships are characterized by infrequent interaction and form between casual acquaintances.
- Result
 - It is more likely that an entrepreneur will get new business ideas through weak-tie rather than strong-tie relationships. WHY?

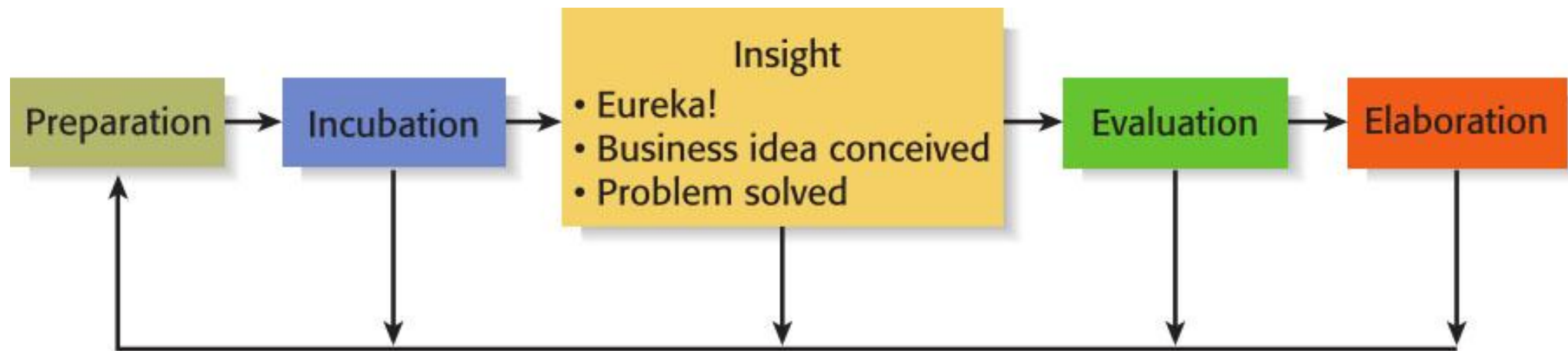
Creativity (1 of 2)

- Creativity
 - Creativity is the process of generating a novel or useful idea.
 - Opportunity recognition may be, at least in part, creative process.



Creativity (2 of 2)

For an individual, the creative process can be broken down into five stages



Techniques for Generating Ideas

Brainstorming

Focus Groups

Library & Internet Research



Other Techniques

Brainstorming

- Is the process of generating several ideas about a specific topic.
- A brainstorming “session” typically involves a group of people, and should be targeted to a specific topic.
- The following are the rules for a brainstorming session.
 - No criticism
 - Freewheeling is encouraged
 - The session should move quickly
 - Leap-frogging is encouraged

Focus Groups

- Focus groups involve a group of people who are familiar with a topic, are brought together to respond to questions, and who are able to shed light on an issue through the give-and-take nature of group discussions.
- They work best as a follow-up to brainstorming, when the general idea for a business has been formulated but further refinement of the idea is needed.



Library and Internet Research (1 of 3)

- Libraries are an often underutilized (غير مستغل) source of information for generating new business ideas.
- The best approach is to talk to a reference librarian, who can point out useful resources, such as industry-specific magazines, trade journals, and industry reports.
- Simply browsing through several issues of a trade journal or an industry report on a topic can spark new ideas.

Library and Internet Research (2 of 3)

Large public and university libraries typically have access to search engines and industry reports that would cost thousands of dollars to access on your own.

Examples of Useful Search Engines & Industry Reports

- BizMiner
- ProQuest
- IBISWorld
- Mintel
- LexisNexis Academic

The screenshot shows the header of the An-Najah National University website. It includes a navigation bar with links for English, social media (Twitter, Facebook, Instagram, YouTube), and email. Below this is a menu with links to 'المراكز العلمية' (Scientific Centers), 'المكتبات' (Libraries), 'نقابة العاملين' (Workers Union), 'العاملون' (Employees), and 'المزيد' (More). The main content area is titled 'قواعد البيانات العلمية' (Scientific Databases) and contains a paragraph in Arabic: 'لدى المكتبة مجموعة كبيرة من الكتب الإلكترونية، المجالات والعديد من الخدمات التي تساعد الطلبة والموظفين في مجال البحث العلمي واداء الوظائف المطلوبة منهم.' Below this is a section titled 'قواعد بيانات باللغة الإنجليزية' (Scientific Databases in English) which lists: ScienceDirect, SpringerLink, (Social Science Research Network (SSRN), and Scientific Research.

English | | | | | البريد | زاجل

المراكز العلمية | المكتبات | نقابة العاملين | العاملون | المزيد

جامعة النجاح الوطنية
An-Najah National University

قواعد البيانات العلمية

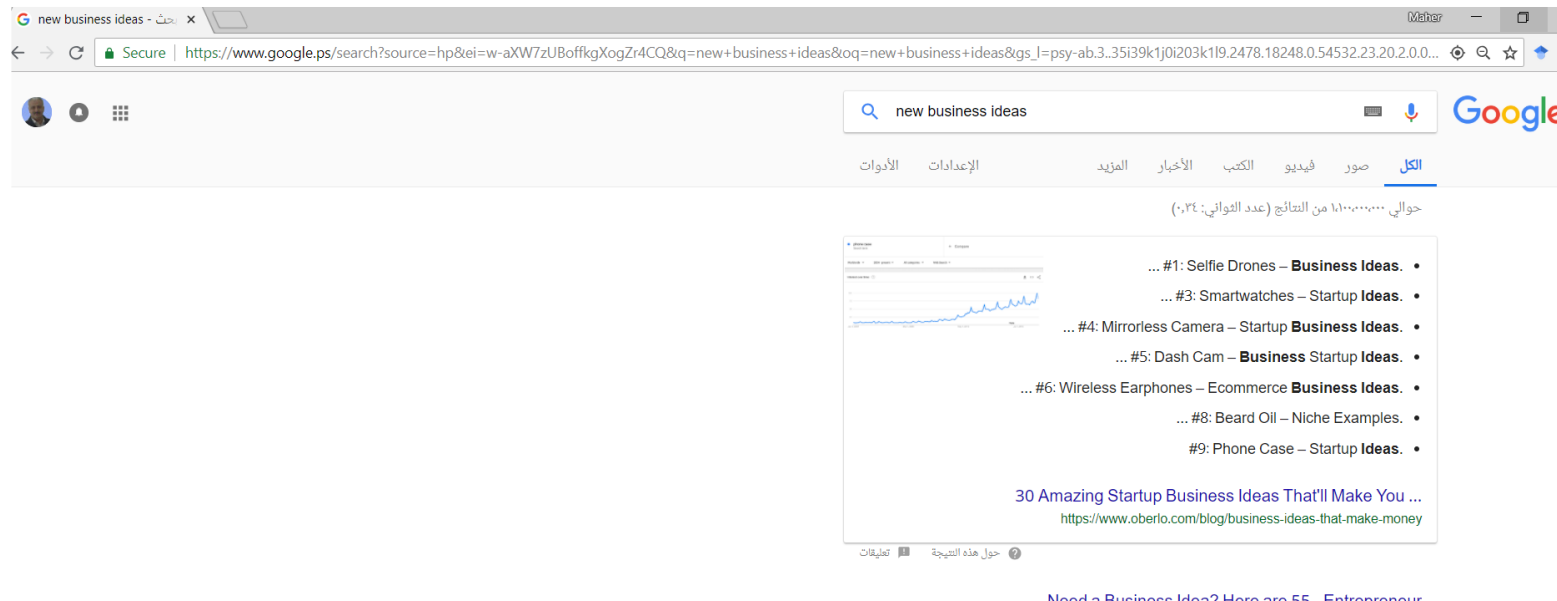
لدى المكتبة مجموعة كبيرة من الكتب الإلكترونية، المجالات والعديد من الخدمات التي تساعد الطلبة والموظفين في مجال البحث العلمي واداء الوظائف المطلوبة منهم.

قواعد بيانات باللغة الإنجليزية

- ScienceDirect
- SpringerLink
- (Social Science Research Network (SSRN
- Scientific Research

Library and Internet Research (3 of 3)

- Internet Research
 - If you are starting from scratch, simply typing “**new business ideas**” into a search engine will produce links to newspaper and magazine articles about the “hottest” and “latest” new business ideas.



Other Techniques (1 of 5)

- Customer Advisory Boards

- Some companies set up customer advisory boards that meet regularly to discuss needs, wants, and problems that may lead to new ideas.

“A Customer Advisory Board (CAB) is a marketing program made up of strategic customers who work closely with company executives to provide guidance on corporate strategies, offer input on products and services, and address and create solutions to industry challenges.”



Eyal Danon

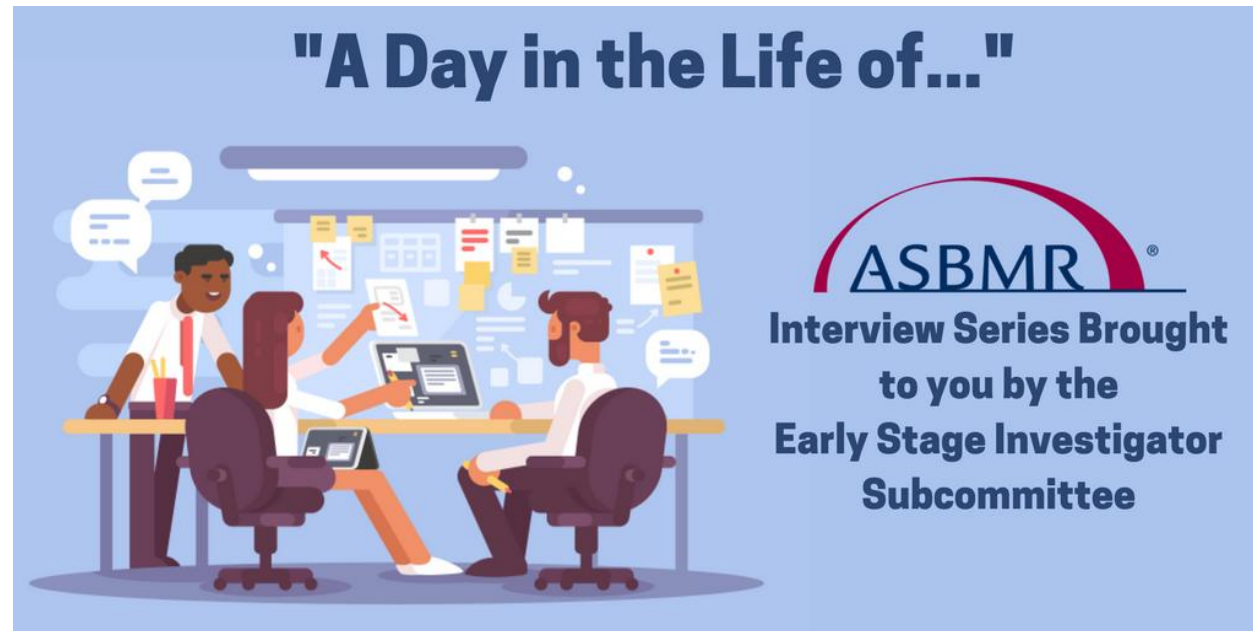
Founder and CEO, Ignite Advisory Group

Ignite’s proven methodology and process for managing and evolving customer advisory boards has been developed from completing over 300 engagements for leading B2B companies.

Other Techniques (2 of 5)

- **Day-In-The-Life Research**

- A type of anthropological research, where the employees of a company spend a day with a customer.



Other Techniques (3 of 5)

- **SCAMPER**

- Innovation is about improving existing processes. At its core SCAMPER is an easy to implement creative thinking and problem solving method that facilitates continuous improvement.
- There is a quote from Thomas Edison that goes “**There’s a way to do it better – find it.**” This is exactly what innovation is about, it’s about improving existing processes and concepts instead of creating something entirely new. You don’t have to reinvent the wheel, but you can make it run more efficient. The SCAMPER method is built on this understanding.

What is SCAMPER?

SCAMPER is actually an acronym and each letter stands for one thinking technique: **S**ubstitute, **C**ombine, **A**dapt, **M**odify, **P**ut to another use, **E**liminate, **R**earrange.

So SCAMPER is a collection of these various techniques and you can choose to use one or all of the seven methods to **find solutions to your problems**. It's also possible to jump from one technique to another.

The SCAMPER method is very similar to Design Thinking in that both these concepts aim to find solutions to problems. However, Design Thinking places the human factor at the center and its goal is to find creative ways to solve problems. The SCAMPER technique is more focused on the process of finding unusual and creative solutions to problems, but also to come up with innovative ideas, and the goal of improving a product or service.



Eliminate

Think of what might happen if you eliminated or parts of your idea. Simplify, reduce or eliminate components. Through repeated trimming of ideas, objects, and processes, you can gradually narrow your challenge down to that part or function that is most important.

How can I simplify it?
What parts can be removed without altering its function?
What's non-essential or unnecessary?
Can the rules be eliminated?
What feature can I omit?



Put to Other Uses

Think of how you might be able to put your current idea to other uses, or think of what you could reuse from somewhere else in order to solve your problem. Many times, an idea only becomes great when applied differently than first imagined.

What else can it be used for?

Can it be used by people other than those it was originally intended for?
Are there other possible uses if it's modified?



Modify/Magnify /Minify

Think about changing part or all of the current situation, or to distort it in an unusual way.

How could it be made larger, stronger, thicker, smaller, lighter or shorter?
Can I change the shape or give it a new twist?
What can be exaggerated or overstated?
Can I increase its frequency?



Rearrange (or Reverse)

Think of what you would do if part of your problem, product or process worked in reverse or were done in a different order.

Can I interchange components?
Are there other patterns, layouts or sequences I can use?
Should I turn it around? Up instead of down? Down instead of up?
What if I consider it backwards?
What if I try doing the exact opposite of what I originally intended?



Substitute

Think about replacing part of the problem, product or process with something else. By looking for replacements you can often come up with new ideas. You can change things, places, procedures, people, ideas, and even emotions.

Can I replace or change any parts?
Can I replace someone involved?
Can the rules be changed?
Can I use other ingredients or materials?
Can I use other processes or procedures?
Can I change its color, roughness, sound or smell?
Can I substitute one part for another?
Can I change my feelings or attitude towards it?

SCAMPER MINDMAP



Jumpstart 5.3



Adapt

Think about adapting an existing idea to solve your problem. Bear in mind that many new ideas or inventions are borrowed to some degree.

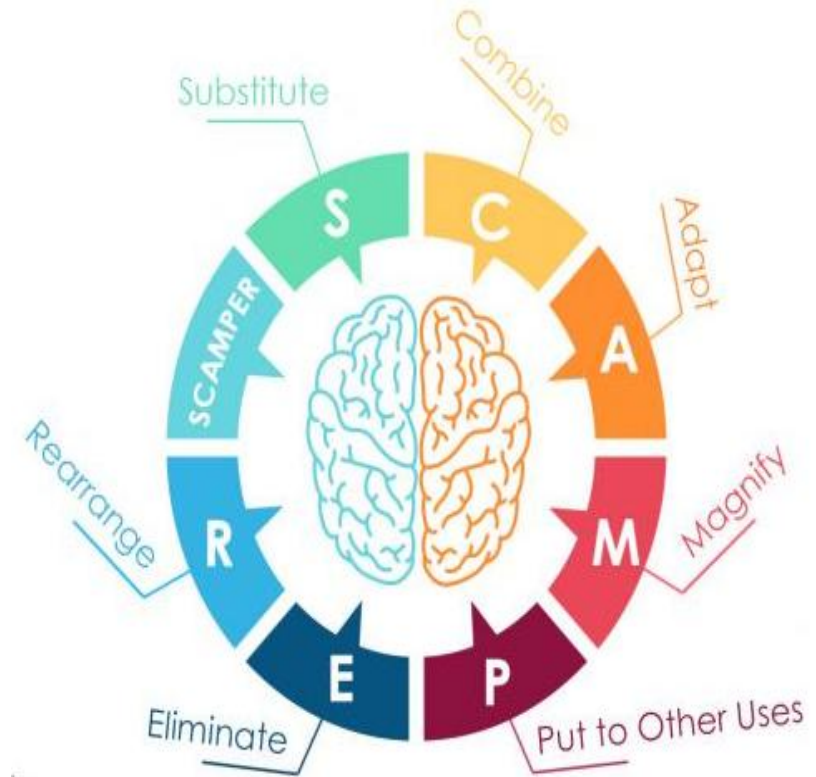
Which parts of the product/ processes could be changed to suit a special condition or purpose?
Does the past offer any lessons with similar ideas?
What could I copy, borrow or steal?
Whom could I emulate?
What ideas could I incorporate?
What processes can be adapted?
What different contexts can I put my concept in?
What ideas outside my field can I incorporate?



Combine

Think about combining two or more parts of your problem to create a different product or process or to enhance their synergy. A great deal of creative thinking involves combining previously unrelated ideas, goods, or services to create something new.

What ideas or parts can be combined?
Can I combine or merge it with other objects?
What can be combined to maximize the number of uses?
What materials could be combined?
Can I combine different talents to improve it?



Substitute



Substitute Example

SCAMPER – Tool for Innovation

↓

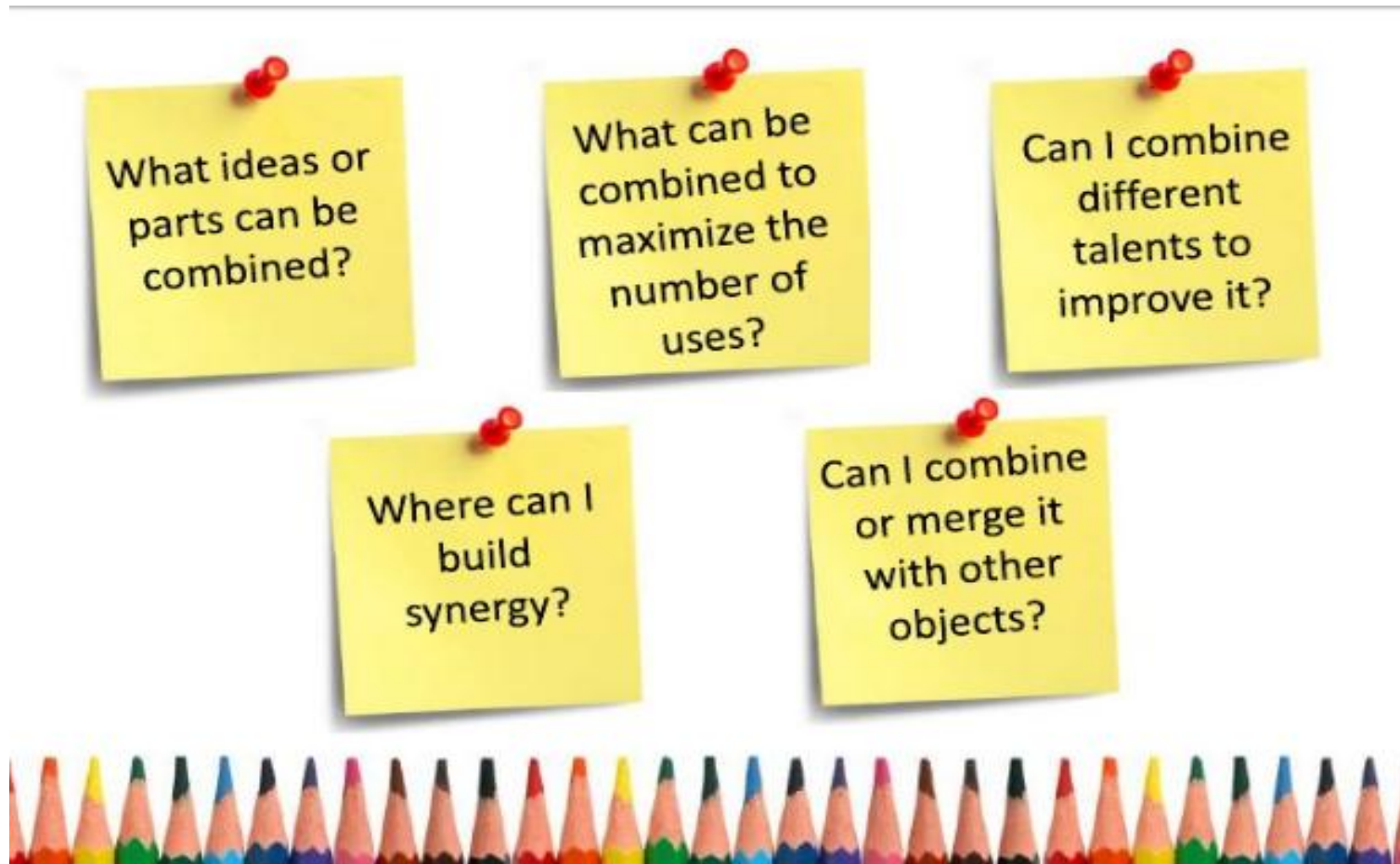
Substitute

Kerosene with
solar power



The collage illustrates the 'Substitute' principle of the SCAMPER tool. It shows a traditional kerosene lantern being replaced by a solar-powered water purifier. The top part of the collage features the text 'SCAMPER – Tool for Innovation' with a red arrow pointing down to the word 'Substitute'. Below this, the text 'Kerosene with solar power' is displayed. The bottom part of the collage contains three images: a traditional kerosene lantern, a man holding a blue and white solar-powered water purifier, and a close-up of the solar-powered device's components.

Combine



Combine Example



Adapt

Is there
something
similar to it, but
in a different
context?

What could I
borrow ?

Whom could
I emulate?

What ideas
outside my
field can I
incorporate?

What
different
contexts can I
put my
concept in?



Adapt Example

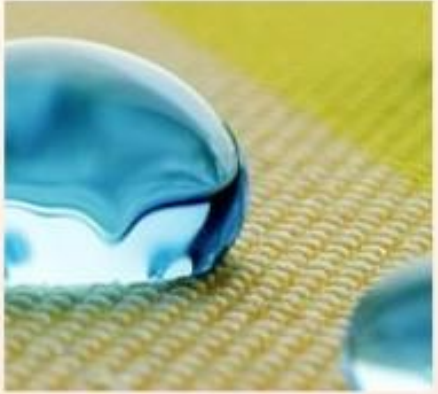
SCAMPER – Tool for Innovation



Adapt

The "lotus effect," a dirt repellant technique adapted from the lotus plant

On textiles treated with BASF Mincor®, billions of nanoparticles are packed closely, and do not let in even a speck of dust.



Modify

What can be
magnified or
made larger?

What can be
exaggerated
or
overstated?

Can I
increase its
frequency?

Can I make
multiple
copies?

Can I add
extra features
or somehow
add extra
value?



Modify Example



SCAMPER



Minify Wheelchair

- Can be folded up and taken by vehicles easily

Modify Usage

- Force sensors give added power to the wheels
- Conceal when not used

Minify (or Minimize)



Put to another use

What else can it
be used for?

How would
children or
older people
use it?

Are there
other
possible
uses if it's
modified?

Are there new
ways to use it
in its current
shape or
form?

How would
people with
different
disabilities
use it?



Put to another use Example

SCAMPER – Tool for Innovation

↓

Put to other uses

- car tyres used as swings

A photograph of a young girl with curly hair, wearing a red dress over a white shirt, sitting on a swing. The swing seat is made from a red car tire, and the chains are attached to a metal frame. The background shows a grassy area and a fence.

SCAMPER

↓

Put to other use

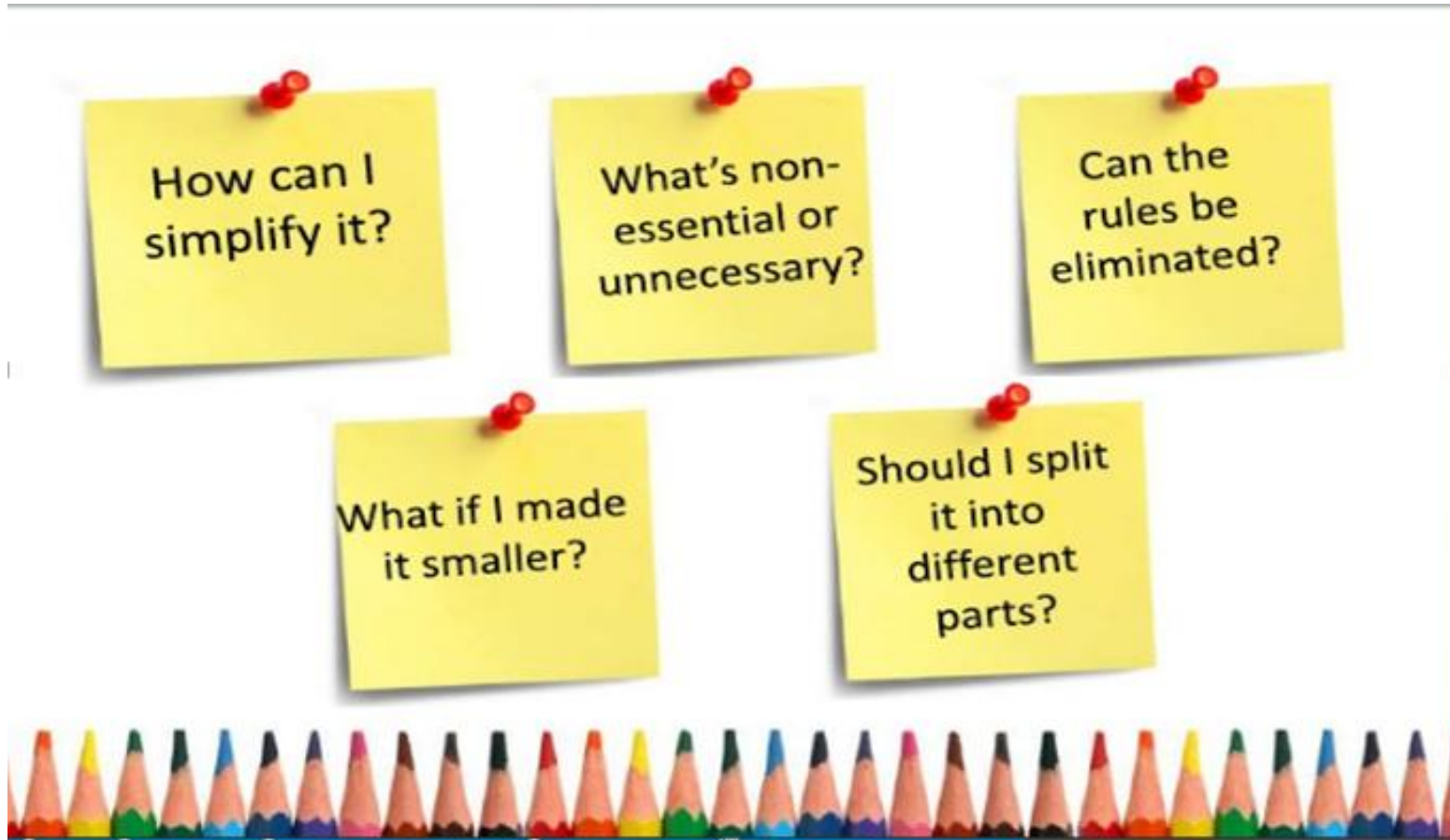
Makes a comfortable reading couch.
Includes seating for three!

A photograph of a black, rectangular couch made from a recycled refrigerator. The couch has a curved backrest and is mounted on four small wheels. A red reading lamp is attached to the backrest. The couch is shown in a room with a white wall and a wooden floor. The text 'Furniture from old fridge' is written on the left side of the image.

Furniture from old fridge

<http://inhabitat.com/transfodesign-turns-a-recycled-refrigerator-into-a-cool-couch/>

Eliminate

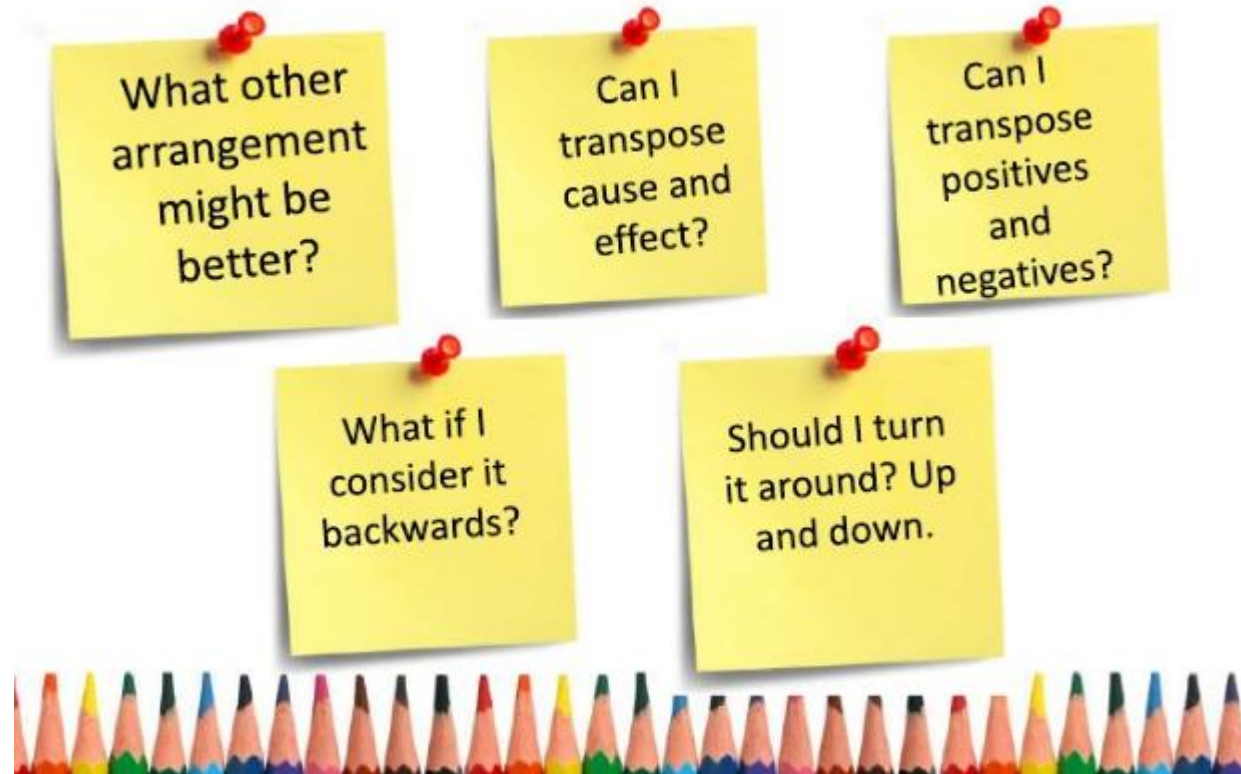


Eliminate Example

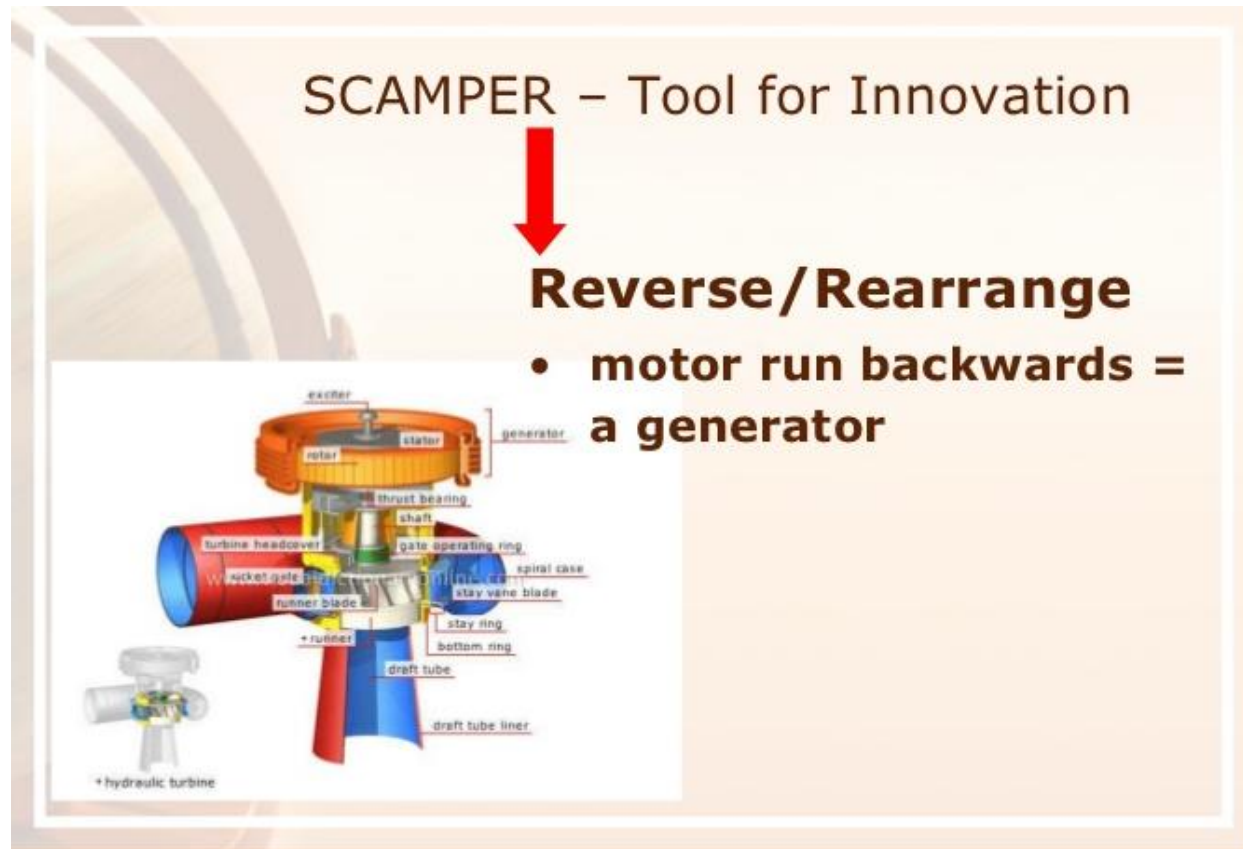
ELIMINATE: ANTENNA AND BASE STATION DIAL PAD



Reverse (Rearrange)



Reserve Example



Why we need SCAMPER

SCAMPER can be used either in person or in group brainstorming sessions (in which there are time constraints and lack of innovative ideas)

DO NOT say “DON’T know” Try your best to find out the answer for each questions by SCAMPERS ,This is your problem, your job, your work, your opportunity so you should know which are principle questions



Apply the SCAMPER technique on:



Other Techniques (4 of 5)

- **Six Thinking Hats**

- Six Thinking Hats or Edward de Bono's Six Hats is a good decision making technique and method for group discussions and individual thinking. Combined with the parallel thinking process, this technique helps groups think more effectively. It is a means to organize thinking processes in a detailed and cohesive manner.

Six Thinking Hats

COLOURED HAT	THINK OF	DETAILED DESCRIPTION
	White paper	The white hat is about data and information . It is used to record information that is currently available and to identify further information that may be needed.
	Fire and warmth	The red hat is associated with feelings, intuition, and emotion . The red hat allows people to put forward feelings without justification or prejudice.
	Sunshine	The yellow hat is for a positive view of things . It looks for benefits in a situation. This hat encourages a positive view even in people who are always critical.
	A stern judge	The black hat relates to caution . It is used for critical judgement. Sometimes it is easy to overuse the black hat.
	Vegetation and rich growth	The green hat is for creative thinking and generating new ideas . This is your creative thinking cap.
	The sky and overview	The blue hat is about process control . It is used for thinking about thinking. The blue hat asks for summaries, conclusions and decisions.

What is parallel thinking ???

At any moment everyone is looking in the same direction



Relationship between 6 Hats & Parallel thinking

Everybody shares each other's opinions about the problems, advantages, facts, reducing distraction and supporting thought cross pollination. This will be accomplished because everyone will put on a hat together, for instance the white hat. After the attendants have expressed their thoughts in a round of discussion, they will put on the next hat. In this way all the attendants will think in the same way at the same time.

THE SIX THINKING HATS

Source: Six Thinking Hats, Dr. Edward de Bono



BLUE HAT

THE CONDUCTOR



WHITE HAT

THE VOICE OF REASON



RED HAT

THE INSTINCTS



YELLOW HAT

THE OPTIMIST



BLACK HAT

THE DEVIL'S ADVOCATE



GREEN HAT

THE CREATIVE THINKER

Exercise (using the 6 hats in decision making)

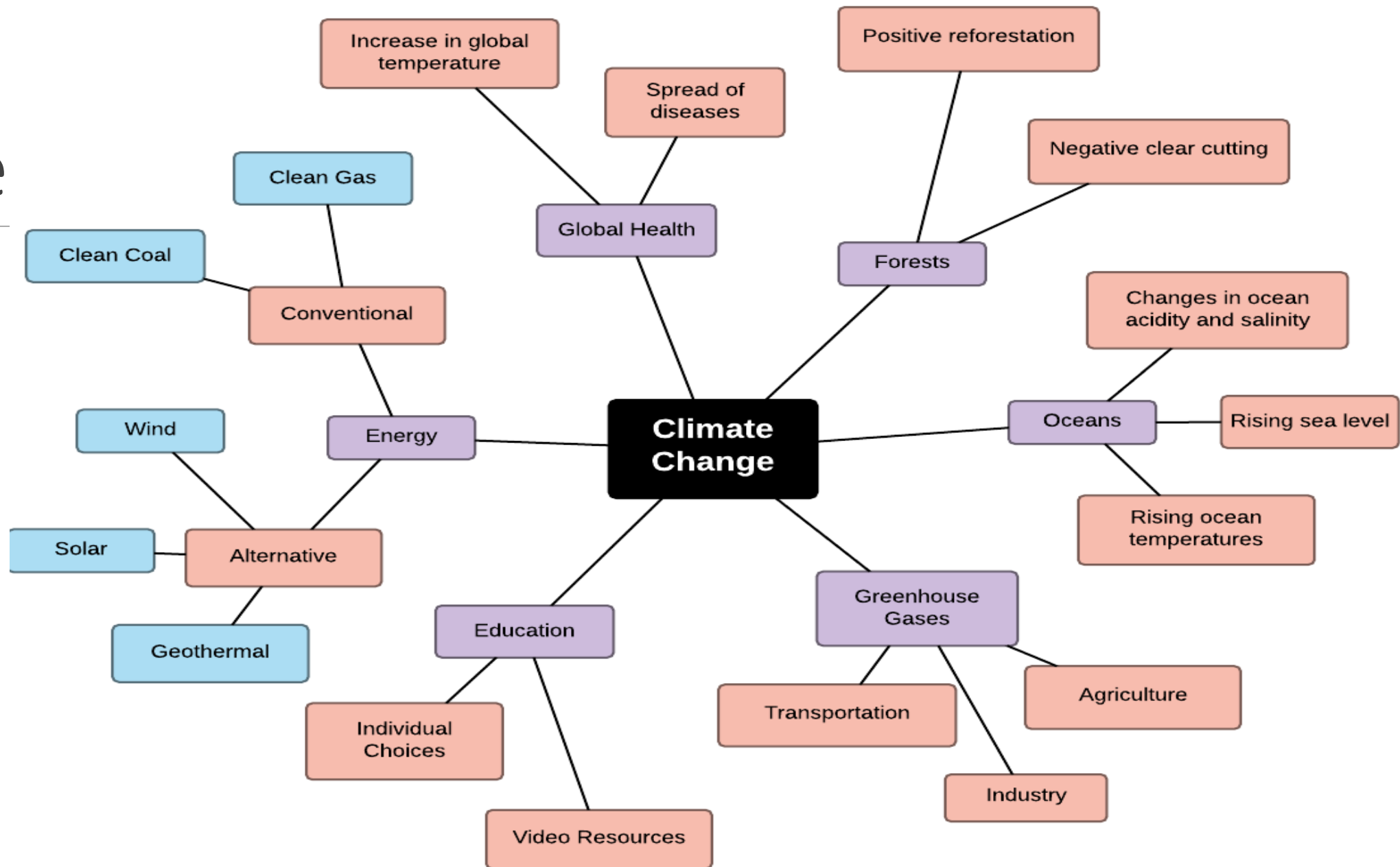
Someone suggested adding salt to the ice cream, read the comments and type the hat type thinking in the table?



Other Techniques (5 of 5)

- **Mind mapping**
 - Mind maps, developed by Tony Buzan are an effective method of note-taking and useful for the generation of ideas by associations
 - To make a mind map, one starts in the center of the page with the main idea, and works outward in all directions, producing a growing and organized structure composed of key words and key images

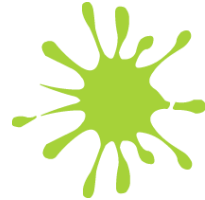
Example



Issues you should know.....



Reasons behind people buying a product or service



Ideas Generation Strategies:

- Identify gaps in the local market
- Develop an existing business model
- Solve a problem-Resolve a common complaint
- Take advantage of the recent technology and market trends
- Take advantage of your strengths, skills and experience



Ideas screening

Business Idea Screening

- Is this business idea likely has a promising financial return?
- Do I have the skills to manage this work? If I do not have it, can I employ people to learn or have those skills?
- Will I be able to raise the money/fund to start this proposed business?
- Am I able to manage this business?
- Will I enjoy working in this environment?

Choosing the best Idea



When choosing your business idea, make sure the following:

- ☐ You will enjoy this work, otherwise you will have difficulty maintaining your motivation on bad days
- ☐ Be realistic, Know your capabilities , and the points you will need help with. Remember, you do not need to know everything, you can take free advice/mentor often or pay for expert opinion.